Linear regression. Linear regression models the target as a weighted sum of input features plus a bias term. The weights are usually chosen to minimize the mean squared error between predictions and true values. For a single feature, the closed-form least-squares solution gives the slope as the covariance of x and y divided by the variance of x. When there are many features, the same problem is solved with the normal equation or, for large datasets, with gradient descent. Gradient descent repeatedly moves the weights in the direction opposite to the gradient of the loss. The learning rate controls the size of each step. A learning rate that is too large can cause the loss to diverge, while one that is too small makes training slow. Feature scaling, such as standardizing each feature to zero mean and unit variance, often makes gradient descent converge much faster.

Logistic regression. Logistic regression is a classification method. It computes a linear score and passes it through the sigmoid function, which maps any real number into the interval between zero and one. The output is read as the probability that the example belongs to the positive class. Because there is no closed-form solution, the weights are learned by maximizing the likelihood, which is equivalent to minimizing the binary cross-entropy loss. Cross-entropy penalizes confident wrong predictions heavily, so it is preferred over squared error for classification. For problems with more than two classes, the sigmoid is replaced by the softmax function, which converts a vector of scores into a probability distribution that sums to one. Regularization adds a penalty on the size of the weights. L2 regularization, also called ridge, penalizes the sum of squared weights, while L1 regularization penalizes their absolute values and can drive some weights to exactly zero.